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'Table 1 shows that…': A local grammar of graphic data commentary in discourse of Economics

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ABSTRACT

This paper analyzes discourse acts in an ESP context from a local grammar perspective. We illustrate this analysis by constructing a local grammar for the discourse act of commenting on information in graphics, i.e., graphic data commentary in the discourse of Economics. Based on a corpus of research articles of Economics, we extracted instances of graphic data commentary that contain lexical items indicating different types of graphics (e.g., table, figure). Eight functional terminologies (e.g., 'Graphic', 'Data') were identified from the instances for the subsequent local grammar analysis, which in turn led to 17 frequently occurring semantic patterns, i.e., local grammar patterns. Each pattern has been observed to have a distinctive way to organize meanings and a corresponding communicative function, fitting the variation of the context. The study indicates that local grammar can provide a precise and systematic description of the lexical-semantic means that economists rely on to communicate graphic information and can offer pedagogical insights into the teaching of academic writing in an ESP context.

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1. Introduction

Texts of a particular discipline or genre are often characterized by the performing of certain discourse acts. For example, the discourse act of citation can be a distinctive feature of academic discourse (Hyland, 1999), and the act of defining characterizes the text of dictionaries (Barnbrook, 2002). Some discourse acts tend to be more heavily conducted in the discourse of a certain discipline. For example, exemplification is observed to be more frequently performed in academic texts of soft sciences than those of hard sciences (Hyland, 2007). Due to the abundance of economic data, the discourse of Economics is "particularly dependent on the use of the graphic image" (Duplass, 1996, p. 33) and is "more of a discussion and analysis of facts and figures" (Bhatia, 2010, p. 41). To facilitate the understanding of abstract economic concepts and their application, economists need to frequently perform the discourse act of commentary of graphic data. Given the importance of this discourse act in the discourse of Economics, this study sets out to analyze how such an act is phraseologically and semantically realized from a local grammar perspective so as to explore the characteristic way of language use in the discourse of Economics and obtain new insights for teaching academic writing in a particular discipline.

In this study, we use the umbrella term 'graphic data commentary' to refer to the act of introducing and discussing the graphics (including tables, figures, charts, graphs, diagrams, etc.) for data presentation. This terming was inspired by Swales

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and Feak's (1994) 'data commentary' which refers to the rhetoric means in academic writing by which scholars visually display research evidence, communicate data, and enhance readers' understanding of knowledge (p. 77). Compared with Swales and Feak's (1994) data commentary, our definition of graphic data commentary is much narrower, focusing on only the data presented explicitly in graphic forms. In other words, only the instances of graphic data commentary that contain words of graphics (e.g., table, figure, graph) were included in our analysis. In consequence, our definition of graphic data commentary corresponds more closely to a sub-type of Swales and Feak's (1994) data commentary, i.e., "pointing out the location of data and summarizing the graphic content" (p. 80). We acknowledge that focusing on these language instances makes it impossible to exhaust the linguistic expressions of graphic data commentary since many expressions are not necessarily associated with particular words of graphics. However, concentration on the most explicitly expressed instances enables us to deal with the most recognizable and typical expressions of graphic data commentary and we are confident that a considerable number of instances can be obtained for this exploratory study.

To date, many studies (e.g., Clymo, 2014; Grech, 2019; McDonald, 2001; Saver, 2006) have been conducted to address graphic data, the purpose of which is to instruct learners on how to design and present graphics in a scientific way. Linguists, on the other hand, see the act of writing about data in graphics as a rhetorical means of information transfer (Widdowson, 1979) or data commentary (Swales and Feak, 1994, 1997) in scientific writing or as a phraseological behavior associated with reporting verbs (e.g., Francis et al., 1996; Yang and Wei, 2017).

It is clear that most studies on graphic data commentary have focused on its stylistic, rhetorical, or phraseological features. However, to our knowledge, no study has yet investigated the linguistic realization of this discourse act from a lexical-semantic perspective, especially in the discourse of a particular discipline. Since the ultimate intention of using stylistic, rhetorical, or phraseological means of graphic data commentary is to communicate meanings to the readers, exploration of the semantic means to realize this discourse act thus is of high research value. To this end, we draw on the local grammar approach (Hunston and Sinclair, 2000) in corpus linguistics, as recent research has shown that local grammar provides a new perspective and more useful means to describe discourse acts in different text types (e.g., Cheng and Ching, 2018; Hunston and Su, 2019; Su and Zhang, 2020; Su et al., 2021). As "a grammar of a discourse function" (Hunston and Su, 2019, p. 571), local grammar concentrates on the description of the linguistic realization of a specific communicative function or discourse act from a lexical-semantic perspective, which we think can be more useful for investigating graphic data commentary. Pedagogically, the local grammar analysis of this particular discourse act can provide new insights for the design of ESP writing courses, as will be elaborated in Section 7.

Drawing on insights from local grammar, this paper analyzes how graphic data commentary in the discourse of Economics is linguistically realized by lexical-semantic means and thereupon explores the methodological and pedagogical value of local grammar in ESP research. It is also argued that a local grammar approach can enrich corpus-based ESP/EAP genre research (e.g., Lu et al., 2021; Moreno and Swales, 2018) by facilitating the exploration of the linguistic features, especially lexical-semantic features of discourse acts or functions that characterize particular rhetorical moves and steps in academic writing of a particular discipline.

2. Review of research on graphic data commentary

The discourse act of graphic data commentary is widely performed in the scientific writing of a large number of disciplines, including Geography, Economics, among many others (Duplass, 1996). By using graphics, scientific writers can 1) reduce reading time by summarizing key information, 2) supplement the main text to clarify complex information, 3) add visual effect to text to enhance interpretability of knowledge, and 4) reduce word counts to save space (Clymo, 2014; Franzblau and Chung, 2012; Saver, 2006). On the other hand, how to effectively write about the data presented in these graphics might be a more challenging yet critical task in scientific writing (Swales and Feak, 1994, 1997; Widdowson, 1979). Given this, researchers have been engaged in providing guidance for the appropriate use of graphics to increase their readability and exploring effective rhetorical and phraseological resources to discuss the information in them.

2.1. Research on the design of standardized graphics

Many studies (e.g., Annesley, 2010; Clymo, 2014; Durbin, 2004; Franzblau and Chung, 2012; Grech, 2019; Luo and Lu, 2012; McDonald, 2001; Sandusky et al., 2010; Saver, 2006; Vickers et al., 2020) have been conducted focusing on providing standardized ways to design graphics and present data in them. For example, based on the analysis of six typical figures in research articles of Clinical Chemistry, Annesley (2010) listed the attributes of an informative and readable figure or graph, including emphasis on data instead of graph, clarity of data and symbols, and visual balance of axes, to mention only a few (p. 1229). Based on the reading of over 400 research papers of Urology, Vickers et al. (2020) summarized six principles for figure presentation, such as making figures notably improve reader's understanding of study findings and ensuring immediate visual impression of figures. They also provided 15 guidelines for table presentation, like avoiding repeating the results in tables in the text and avoiding reporting *p* values in tables without a careful consideration of the hypothesis. These investigations have provided researchers and students with valuable advice on how to conform to the norms of visual message presentation in scientific writing.

2.2. Rhetoric research

Rhetoric studies on information transfer (Widdowson, 1979), data commentary (Swales and Feak, 1994, 1997, 2012), figure legends (Liu et al., 2023), and commentaries on visuals (Wu et al., 2024) are also closely related to the investigation of graphic data commentary. Based on a case study of the academic writing of four graduate students, Widdowson (1979) points out that in scientific discourse, writers are responsible for transferring the non-verbal information communicated by symbols, formulae, tables, and figures to verbal forms, hence the act of information transfer. By manual reading of research papers drawn from a wide range of disciplines (Swales and Feak, 1994) or by investigating a 80,000-word corpus of research articles in different disciplines (Swales and Feak, 2012), Swales and Feak note that academic writing involves the task in which writers need to discuss data typically displayed in tables, charts or figures. They termed this task 'data commentary' which consists of three sub-tasks to complete: 1) pointing out the location of data and summarizing the graphic content (e.g., *Table 5 shows the most common modes of infection for U.S. business*…), 2) highlighting the information in graphics from the more significant to the less significant (e.g., *As can be seen, in the majority of cases … However, it is alarming to note that …*), and 3) discussing implications, problems, etc. (e.g., *While it may be possible to eliminate hone-to workplace infection … business are still vulnerable to major data loss*…) (Swales and Feak, 1994, p. 80). More critically, it was found that the first type of task is typically realized by a set of lexical-grammatical resources, including a consistent use of the present tense (e.g., *Table 5 shows*, *Figure 4.2 gives*), a preference of a limited set of verbs (e.g. *show*, *present*) and the linking *as*-clause (e.g., *as shown in Table 5*, *as can be seen in Figure 8*) (Swales and Feak, 1994, pp. 80–82). Recently, researchers develop new interest in the rhetorical features of different forms of commentaries on graphic information in academic discourse, such as figure legends (Liu et al., 2023) and commentaries on visuals (Wu et al., 2024). For example, based on a 112,492-word corpus of social science reports, Wu et al. (2024) identified the rhetorical functions (e.g., presenting results, describing/rationalizing experiments) and the typical phrase-frames (e.g., ' … *results are robust to*', '*it has been … that*') of the commentaries on tables and figures. Inspired by the investigation of the lexical-grammatical means of graphic data commentary in the above-mentioned rhetoric studies, the present study further explores other linguistic features, such as discourse-semantic features in the lexical-grammatical resources of graphic data commentary (as we have discussed in Section 1).

2.3. Phraseological research

Corpus linguists are concerned with how graphic data commentary is realized in phraseologies. For example, Francis et al. (1996) observed the phraseological behavior of a small group of reporting verbs, i.e., 'show' group verbs (e.g., *show*, *reveal*, *demonstrate*) in the corpus of Bank of English and noted that these verbs tend to be used particularly for the purpose of indicating facts or situations (p. 101). More importantly, it was further found that in academic discourse, these verbs, though largely used for reporting research findings or observations, are also frequently used to comment on data in various forms of graphics. This phraseological feature of the reporting verbs is even more evident in the academic texts of particular disciplines, such as Materials, as reported in Charles (2006).

Other corpus-based phraseological studies (e.g., Master, 1991; Yang and Wei, 2017), though not explicitly focusing on graphic data commentary, have found that in academic discourse some phraseologies realizing specific collocational patterns can perform the function of commenting on graphic data. For example, based on the observation of a 7,464,109-word corpus consisting of 1,625 published research articles in 23 disciplines, Yang and Wei (2017) found that inanimate subjects indicating figures, tables, values, ratios, etc., frequently co-occur with the 'show' group reporting verbs to form a collocational pattern of '$N_{inanimate}$ V that' (e.g., *Table 1 shows that*, *Figure 3 reveals that*), performing the discourse function of providing explanations of visual data, i.e., graphic data commentary. Yang and Wei (2017) argue that this patterning of language and its discourse function reflect academic writers' use of discourse strategies to provide readers with accurate and clear explanation and interpretation of visual data. Although valuable findings have been obtained, it is still important to note that the above-mentioned studies might have some limitations in terms of the following three aspects. One limitation is that most studies have focused on the collocational or grammatical features of graphic data commentary (e.g., Yang and Wei, 2017). The linguistic realization of this act from a lexical-semantic perspective has not been fully investigated. For example, to our knowledge, no studies have yet explored what specific meanings can be expressed to perform graphic data commentary. Secondly, the methodologies of previous studies are relatively homogenous. Most studies (e.g., Annesley, 2010; Swales and Feak, 2012; Widdowson, 1979) rely on the case study of graphics or the reading of small-sized academic texts to draw observational conclusions. For example, Annesley's (2010) findings are obtained from the study of six figures in research articles of Clinical Chemistry. It is argued that the investigation of the language of graphic data commentary based on large corpora is needed to more systematically explore the form-meaning features of this discourse act. Thirdly, graphic data commentary in a specific academic discipline, such as Economics, deserves closer attention. Previous studies, especially those under rhetoric approach (e.g., Swales and Feak, 1994) and phraseological approach (e.g., Yang and Wei, 2017), tend to deal with graphic data commentary in general academic discourse. It has to be noted that the importance of graphic data commentary varies across disciplines. In academic discourse of Economics, as pointed out in Duplass (1996) and Bhatia (2010), graphic data is of more obvious importance as economic research is characterized by abundance of economic data which requires graphic illustrations for effective explanation. Therefore, restricting the investigation to the discourse of Economics is reasonable, allowing us to concentrate on the most typical text type for graphic data commentary and to deeply explore the

disciplinary characteristics of this discourse act. Inspired by the corpus-driven local grammar research of discourse acts (e.g., Barnbrook, 2002; Cheng and Ching, 2018; Hunston and Sinclair, 2000; Hunston and Su, 2019; Su and Zhang, 2020; Zhang and Su, 2021), we think that the local grammar approach can provide a more systematic means to address graphic data commentary, especially its linguistic realization from a lexical-semantic perspective. In view of this, the present study attempts to use a local grammar approach to explore the characteristic lexical-semantic means to realize graphic data commentary in the discourse of Economics.

3. Local grammar and discourse acts

Local grammar is a meaning-based grammar which describes the discourse-semantic means to express a particular discourse act or function in communication (Su, 2019, p. 53). Each local grammar is defined by a specific discourse act (Su, 2020), providing the semantic patterns (i.e., local grammar patterns) of the language instances that realize such a discourse act. For example, in Hunston and Su's (2019) local grammar of the discourse act of evaluation, instances of evaluation were described as 22 local grammar patterns, such as 'Hinge + Evaluation + Target' (representing instances like *It would be selfish to marry*, *It is awful that it should end like this*, etc.) and 'Evaluator + Hinge + Evaluation + Target' (e.g., *He was happy about people having to move*, *They were keen on the idea of education*, etc.).

This grammar approach has also been found to be useful to facilitate the description of discourse acts in academic discourse and business discourse, such as revealing cause-and-effect relationship (Allen, 2005) and exemplifying (Su and Zhang, 2020; Su et al., 2021; Zhang and Su, 2021) in academic discourse and the act of movement of business data in business discourse (Sui, 2015). For example, Su and Zhang (2020) constructed a local grammar for the discourse act of exemplification in research articles of Linguistics. In the local grammar analysis, Su and Zhang used terminologies specific to exemplification, such as 'Exemplified' (referring to the element being illustrated), 'Exemplification' (referring to the concrete things used to illustrate), and 'Indicator' (referring to the element that signals the act of exemplifying) to analyze phraseologies of exemplification through form-meaning mapping (e.g., the meaning 'Exemplified' is mapped onto the form *These practices include*, as shown in Table 1) into different local grammar patterns.

Table 1
Local grammar analysis of exemplification (Su and Zhang, 2020, p. 5).

Exemplified	Indicator	Exemplification
These practices include	for example	the stance students convey towards their artefacts …
… draw attention to important social features	such as	the language and hierarchical structure of such communities

As Table 1 shows, one group of instances of exemplification which demonstrate a similar way of expressing meanings were analyzed into a common local grammar pattern of 'Exemplified + Indicator + Exemplification'. Other exemplifying expressions which show variations in meaning expression were described as patterns like 'Initiator + Indicator + Exemplification' (e.g., *Through triangulation of, e.g., interview and observational data*) or 'Exemplification + Exemplifying + Exemplified' (e.g., *The following examples illustrate the different types of attempted paraphrases*). These local grammar patterns, together, systematically represent linguists' habitual semantic choices to exemplify knowledge in different communicative contexts.

Unlike previous phraseological studies which reveal the lexical/collocational patterns of reporting verbs for graphic data commentary, the local grammar approach can probe into the semantic properties of this discourse act and systematically reveal the semantic regularities of its lexical/collocational patterns. This deeper exploration is important as the reporting verbs (e.g., *show*, *reveal*) which realize a same lexical/collocational pattern (e.g., $N_{inanimate}$ V that, as found in Yang and Wei, 2017) may manifest different semantic patterns, as shown in Examples (1) and (2).

(1) Fig. 5 shows that they leveled off just after October 14, 2008.
(2) Fig. 1 suggests that the level of earnings management could be related to overall market performance.

As the two examples show, both of them realize the collocational pattern of $N_{inanimate}$ V that. However, this pattern does not reveal the different semantic roles played by *that*-clauses in the two examples: in Example 1, *that they leveled off just after October 14, 2008* points to the data directly observed from a particular graphic, while in Example 2, *that the level of earnings management could be related to overall market performance* gives the writer's interpretation of the graphic data. Focusing on the semantic/functional roles played by the structural elements in phraseologies of a discourse act, local grammar enables a finer-grained description of the form-meaning features of discourse acts. The two *that*-clauses in Examples (1) and (2), in a local grammar analysis, will be differently described as 'Data' and 'Interpretation', revealing the minute semantic choices in the use of the phraseologies.

In light of the precise and context-sensitive description of discourse acts by local grammar, this study attempts to analyze graphic data commentary in the discourse of Economics from a local grammar perspective. We hope our analysis can further demonstrate the descriptive power of local grammar for a new discourse act (other than

evaluation, disclaimer or exemplification) in the ESP discourse and provide more practical insights for the ESP writing pedagogy.

4. Corpus and methodology

4.1. Corpus

Our investigation is based on a self-constructed corpus of published research articles of Economics. Although graphic data commentary is also common in texts of many other disciplines or genres, it has been observed to be more preponderant in the discourse of Economics (e.g., Bhatia, 2010; Duplass, 1996) and academic discourse (e.g., Swales and Feak, 1994, 1997), as discussed above. Therefore, it is safe to say that research articles of Economics can provide a more typical discourse sample to investigate graphic data commentary. The texts we collected are full-length research articles published from the year of 2013–2017 in five international journals of Economics with high impact factor (ranging from 5.9 to 10 in the year of 2023), including *Journal of Retailing*, *Journal of Accounting and Economics*, *Journal of Finance*, *Economic Geography*, and *Journal of Financial Economics*. Texts in the corpus were balanced across both years and journals. For each article, only the body part was collected for the corpus. The abstract, references, and other supplementary parts (e.g., author information, notes, and appendixes) were excluded. The corpus consists of 122 texts, amounting to 982,515 words.

4.2. Working procedure

4.2.1. Identification of lexical items for graphic data commentary

Previous studies (Franzblau and Chung, 2012; Swales and Feak, 1994; Yang and Wei, 2017) and our own observation of the research corpus have shown that making graphic data commentary is explicitly and typically associated with six lexical items indicating different forms of graphics, including *table, figure, diagram, chart, graph*, and *graphic*. Our further reading of 20 randomly selected research articles published in another five leading academic journals of Economics (e.g., *Quarterly Journal of Economics*, *Journal of Economic Perspectives*) confirms the typicality of these lexical items in the discourse of Economics. These lexical items provide the starting point to extract instances of graphic data commentary in the corpus. Although retrieving these words of graphics in the corpus cannot ensure a full coverage of instances of graphic data commentary since expressions of this discourse act are not necessarily associated with particular words of graphics (as pointed out in Section 1), it can capture a considerable number of instances for the current exploratory study and these instances, most typical of graphic data commentary, deserve closer attention. Moreover, focusing on the most explicitly expressed instances via retrieving typical lexical items is a key characteristic of corpus linguistic research.

4.2.2. Data retrieval and identification of local grammar terminologies

We retrieved the six lexical items in the corpus using the concordancer in WordSmith Tools 6.0. In each concordance (e.g., *All controls are winsorized at the 2nd and* 98th *percentiles by fiscal year. Table 2 presents the results of this analysis*), we focused on the local sentence or phraseology containing any one of the six lexical items (e.g., *Table 2 presents the results of this analysis*). When the concordance window was too narrow to display a full sentence/phraseology, wider context was checked for clarification. These instances were subsequently manually checked to filter out those unqualified ones which do not write about graphic information such as *Because many of these table service restaurants (e.g., Chili's, TGI Friday's) feature some very large servings*. From the qualified instances (see Concordance Sample 1 for instance), we generalized the meanings associated with graphic data commentary into a number of functional terminologies.

Concordance Sample 1 Instances of graphic data commentary

Table 4 shows that the results for the VIX VARs are very similar to the others. In particular, there is
Table 2 provides a description of our proxies for potential conflicts of interests for the consultants working for
Table 1 presents basic features of the data. Panel A shows the percentage of sample firms in each CRSP size
Figure 5 plots three additional equilibrium possibilities indexed by p T 0 = 62, 65, and 67. All these time paths
This figure shows the event-time average of cumulative aggregate illiquidity innovations, around a directional

As shown in the sample instances, the initial nouns (e.g., *Table 4*, *Figure 5*) express the common meaning of different forms of graphics. It was therefore further generalized into a more abstract terminology of 'Graphic'. Following this way, we totally identified eight functional terminologies (see Table 2) from the instances.

Table 2
Local grammar terminologies of graphic data commentary.

Terminology	Definition	Example
Graphic	tables, figures, charts, etc. used to illustrate data or relationships	***Table 2*** summarizes the target statistics
Act	The research act conducted for the discussion of graphic data or for other research purposes	The third figure ***displays*** a plot of quarterly realized volatility. We ***plot*** the index in Fig. 2 along with plots of excess returns and volatility
Data	The specific information presented in graphics	The chart shows ***the distribution of the timing of gifts of stock***
Interpretation	Further explanation of the graphic information	Figure 1 suggests ***that the level of earnings management could be related to overall market performance***
Initiator	The initial element that introduces a particular graphic	***According to*** Figure 2, a bubble exists if k/c rises over time
Researcher	The person who reports or analyzes the data in graphics	***We*** find Jensen's alpha is not significantly different from zero in Table 2
Hinge	Element to connect different parts	This is also true ***in*** **Table 2** It had a see-through packaging to show the clear (transparent) cola (***see*** Fig. 4a)
Additional information	Supplementary information for clarification or explanation	See Table 4 and Fig. 1 ***for neighborhood names and locations***

4.2.3. Local grammar analysis

In this step, instances of graphic data commentary were analyzed from a local grammar perspective, the process of which involved mapping the local grammar terminologies in Table 2 onto structural elements of the retrieved instances (as illustrated in Table 3).

Table 3
Pattern 1 'Graphic + Act + Data'.

Graphic	Act	Data
Figure 5	plots	three additional equilibrium possibilities indexed by p T 0 = 62, 65, and 67
Table 1	presents	basic features of the data

As Table 3 shows, two instances from Concordance Sample 1 can be analyzed as follows: The initial noun phrase (e.g., *Figure 5*) indicating a particular table or figure was analyzed as 'Graphic'. The following verb (e.g., *plots*) is used to present the graphic data, thus was termed as 'Act'. The subsequent detailed information (e.g., *three additional equilibrium possibilities*…) matches the terminology of 'Data'. This analysis produced the local grammar pattern of 'Graphic + Act + Data' which represents the semantic regularity of this group of instances.

Following this procedure, we analyzed all the retrieved instances into a set of local grammar patterns, which, combined together, can systematically represent the semantic regularities of graphic data commentary in the discourse of Economics.

5. Results

Due to space limit, we set a frequency threshold for the local grammar patterns identified: only the patterns which have ten or more instances (≥10 times per million words) are reported in the results. This enables us to focus on the most typical semantic patterns of graphic data commentary in the corpus. To more conveniently display the results, we divided the local grammar patterns into five groups according to the initial meaning element in the patterns.

The first group of 691 instances commonly starts with 'Graphic'. Among them, 406 instances were analyzed as the local grammar pattern of 'Graphic + Act + Data' (see Pattern 1 in Table 3). This pattern is characterized by the initial positioning of graphics and a brief introduction of the main content in them. It enables writers to start the discussion of a group of research results through general description of graphic content. In this study, as Pattern 1 has the highest frequency in the corpus, it therefore represents the most typical lexical-semantic means to conduct graphic data commentary in the discourse of Economics. The analysis of the rest 285 instances produced a slightly different pattern 'Graphic + Act + Interpretation' (see Pattern 2 in Table 4). In this pattern, the content following 'Act' does not show the concrete data but gives the economist's understanding of the data, thus was termed as 'Interpretation'. Pattern 2 therefore has the characteristic discourse function of generalizing major findings from a particular graphic.

Table 4
Pattern 2 'Graphic + Act + Interpretation'.

Graphic	Act	Interpretation
Table 4	shows	that the results for the VIX VARs are very similar to the others
Figure 2	confirms	that the relationship exhibits a high degree of consistency over the 52-year study period

812 instances starting with 'Data' were analyzed into seven different local grammar patterns.

In 292 instances (see Table 5), the 'Data' (e.g., *The market is divided into three sections*) is provided first and is linked with the 'Graphic' (e.g., *Fig. 1*) through a verb (e.g., *refer to*) or preposition (e.g., *of*) indicating 'Hinge'. The verbal 'Hinge' inserted in-between functions to draw readers' attention to the target graphic, reflecting writer-reader communication in performing graphic data commentary. Meanwhile, by putting the graphic in parentheses, this pattern can save space for fuller discussion of graphic data.

Table 5
Pattern 3 'Data + Hinge + Graphic'.

Data	Hinge	Graphic
The market is divided into three sections	(refer to	Figure 1)
… the results	of	Table 3

Another 11 instances, with 'Additional information' (e.g., *for wording*) put at the end, realize a pattern 'Data + Hinge + Graphic + Additional information' (see Pattern 4 in Table 6) quite similar to Pattern 3. The addition of the ending prepositional phrase provides the concrete reasons to read a particular graphic.

Table 6
Pattern 4 'Data + Hinge + Graphic + Additional information'.

Data	Hinge	Graphic	Additional information
All items were rated on a seven point semantic-differential scale	(see	Table 1	for wording)
defined by aggregating adjoining zip codes	(see	Table 4 and Fig. 1	for neighborhood names and locations)

165 instances can be analyzed as 'Data + Hinge + Act + Hinge + Graphic' in Table 7. This group of instances shows a passive structure starting with a noun phrase (e.g., *The overall ANOVA results*) indicating 'Data''. The link verb (e.g., *are*) plays the role of 'Hinge', linking the noun with another verb realizing 'Act' (e.g., *provided*). The source of the data is pointed out with a prepositional phrase realizing 'Hinge' and 'Graphic' (e.g., *in Table 2*). The resultant Pattern 5 is suitable for the context in which a set of data has been mentioned but still needs further visual elaboration.

Table 7
Pattern 5 'Data + Hinge + Act + Hinge + Graphic'.

Data	Hinge	Act	Hinge	Graphic
The overall ANOVA results	are	provided	in	Table 2
The actual and predicted sales shares	are	shown	in	Figure 3

In 113 instances (as Pattern 6 in Table 8), the initial sentence which gives a long description of graphic information (e.g., *The PCAOB reports in our sample were issued…*) provides 'Data'. The parenthesized graphic at the end (e.g., *(Table 2)*) signals the location of the commentary. With graphic name put in parentheses, Patterns 6 saves space for a detailed data analysis, rather suitable for an in-depth discussion of graphic information.

Table 8
Pattern 6 'Data + Graphic'.

Data	Graphic
The PCAOB reports in our sample were issued between January 1, 2005 and December 31, 2007	(Table 2)
12 were located in Houston, Texas, 4 in Sugar Land, Texas, 1 in Missouri City, Texas and Fresno, Texas	(Fig. 1)

Pattern 7 which covers 103 instances (see Table 9) demonstrates a more complicated arrangement of meanings. The 'Data' in a particular 'Graphic' is interpreted (e.g., *consistent with either explanation*). This pattern 'Data + Hinge + Graphic + Hinge + Interpretation' can be used to comment on the value of concrete sets of data in graphics.

Table 9
Pattern 7 'Data + Hinge + Graphic + Hinge + Interpretation'.

Data	Hinge	Graphic	Hinge	Interpretation
The post-gift date stock price declines reported	in	Figure 4 and Table 3	are	consistent with either explanation
The evidence	in	Figure 1	is	similar to the findings of Fujimoto (2004)

As an extension of Pattern 3, Pattern 8 (see Table 10) which covers 65 instances involves two additional elements of 'Act' and 'Data' to form a more complicated 'Data + Hinge + Graphic + Act + Data'. Similarly, Pattern 9 'Data + Hinge + Graphic + Act + Interpretation' which covers 63 instances (see Table 11) extends Pattern 3 by adding 'Act' and 'Interpretation'. These two patterns can be used to precisely analyze or further elaborate on a particular group of data in graphics.

Table 10
Pattern 8 'Data + Hinge + Graphic + Act + Data'.

Data	Hinge	Graphic	Act	Data
The model conceptualized	in	Figure 1	yielded	a good overall fit
The information	in	Figure 1	provides	a method for selecting a threshold θ

Table 11
Pattern 9 'Data + Hinge + Graphic + Act + Interpretation'.

Data	Hinge	Graphic	Act	Interpretation
The dotted line in the top row of plots	in	Figure 4	indicates	that the federal funds rate trends down during most of the extended event period
Our correlations	in	Table 7	show	that better governed countries have more developed equity markets

The third group of 192 instances similarly starts with a preposition realizing 'Initiator' (e.g., *in*, *as*, *from*) whose common function is to direct readers' attention to a specific graphic.

In 95 instances (see Pattern 10 in Table 12), the first element 'Initiator', realized by a preposition (e.g., *from*) or prepositional phrase (e.g., *as can be seen in*), brings out the 'Graphic' (e.g., *Fig. 5*). The clause at the end (e.g., *the retailer's actual assortment yields profits*…) provides concrete description of the graphic data, thus termed as 'Data'. Another ten instances, analyzed as Pattern 11 in Table 13, manifest a variant of Pattern 10. The only difference lies in the addition of 'Act' between 'Graphic' and 'Data'. Both Pattern 10 and Pattern 11 are observed to frequently occur at the beginning of a paragraph, initiating a detailed analysis of the data in a particular graphic or presenting graphic evidence to support an argument.

Table 12
Pattern 10 'Initiator + Graphic + Data'.

Initiator	Graphic	Data
As can be seen in	Figure 5,	the retailer's actual assortment yields profits well below the efficient frontier
From	Table 1,	the pooled correlation coefficient for the two measures is −0.95

Table 13
Pattern 11 'Initiator + Graphic + Act + Data'.

Initiator	Graphic	Act	Data
As	Figure 3	shows,	participants in the assessment prime condition assigned more value to the frozen yogurt
As	Table 3	indicates,	the median bank funds 56 % of its assets with deposits

87 instances can be analyzed as two similar patterns, 65 of them as Pattern 12 (Table 14) and the rest 22 as Pattern 13 (Table 15). In Pattern 12, the 'Initiator' and 'Graphic' (e.g., *In Fig. 5*) point to the graphic under discussion. The subsequent clause consists of three semantic elements: 'Researcher' realized by a pronoun (e.g., *we*), 'Act' by a verb (e.g., *compute*), and 'Data' by a noun phrase (e.g., *the profit curve for optimal MNL assortments*…). Pattern 13 is slightly different, with 'Act' followed by a *that*-clause (e.g., *that the coefficient on a fund's own friendliness is*…) indicating interpretation of the graphic data. By showing the intention to analyze or interpret a particular group of graphic data, these two patterns enable economists to introduce research process using graphic data or invite readers to explain the graphic data together.

Table 14
Pattern 12 'Initiator + Graphic + Researcher + Act + Data'.

Initiator	Graphic	Researcher	Act	Data
In	Figure 5	we	compute	the profit curve for optimal MNL assortments of size 1–50 DVD players
In	Table 2, Panel B	we	highlight	the mean, median, and variance of raw weekly returns for three subsamples

Table 15
Pattern 13 'Initiator + Graphic + Researcher + Act + Interpretation'.

Initiator	Graphic	Researcher	Act	Interpretation
In	Tables 1 and 2,	we	found	that the combination $\alpha = \lambda = 0.95$ led to the best forecast performance
From	Table 8,	we	can see	that the coefficient on a fund's own friendliness is remarkably stable

The fourth group of three patterns commonly starts with 'Interpretation', covering 86 instances.

48 instances realize Pattern 14 'Interpretation + Graphic' (see Table 16). Slightly different from Pattern 6, the content placed before the parentheses (e.g., *The model specification is the same as the full model in earlier calculations*) does not directly show the graphic data but provides explanation of it, thus was termed as 'Interpretation'. Pattern 14 allows a full elaboration on data thanks to the parenthesized graphics.

Table 16
Pattern 14 'Interpretation + Graphic'.

Interpretation	Graphic
The model specification is the same as the full model in earlier calculations	(Table 2)
Firms that state a strong need for external capital in the future have larger lines of credit, contrary to our findings for non-operational cash	(Table 6)

Table 17 shows the analysis of 22 instances which produced Pattern 15 'Interpretation + Hinge + Graphic'. This pattern is characterized by evaluative interpretation of data at the beginning (e.g., *This is also true*). The subsequent prepositional phrase (e.g., *in Table 2*) points out the corresponding graphic of the interpretation. The evaluation embedded in interpretation enables Pattern 15 to identify valuable features in the graphic data.

Table 17
Pattern 15 'Interpretation + Hinge + Graphic'.

Interpretation	Hinge	Graphic
This is also true	in	Table 2
The merger waves of the late 1980s and late 1990s are evident	in	the table

In 16 instances (see Table 18), the initial 'Interpretation' is followed by a semi-fixed phrase (e.g., *as suggested in Table 1*): the preposition *as* realizes 'Hinge', the past participle of a reporting verb (e.g., *suggested*) indicates 'Act', and a prepositional phrase (e.g., *in Table 1*) indicates 'Hinge' and 'Graphic'. Similar to Pattern 14, this Pattern 16 is also suitable for a full elaboration on graphic information.

Table 18
Pattern 16 'Interpretation + Hinge + Act + Hinge + Graphic'.

Interpretation	Hinge	Act	Hinge	Graphic
Thematic designs within stores could serve as attracting or avoidance destinations,	as	suggested	in	Table 1
This is the source of the differences in time-series statistics across cases,	as	reported	in	Table 7

74 instances which commonly start with 'Researcher' (e.g., *we*) form the last pattern group. They were all analyzed as Pattern 17 'Researcher + Act + Data + Hinge + Graphic' (see Table 19). As a variant of Pattern 12, Pattern 17 relocates the initial prepositional phrase (e.g., *in Fig. 4*) to the end of the expression. This allows economists to more naturally link the graphic data commentary with preceding research activities.

Table 19
Pattern 17 'Researcher + Act + Data + Hinge + Graphic'.

Researcher	Act	Data	Hinge	Graphic
We	plot	the time series of weekly estimates of the coefficient of the ARS indicator variable	in	Figure 4
We	find	Jensen's alpha is not significantly different from zero	in	Table 2

6. Discussion

We collected all the 17 patterns (ranked in frequency order) in Table 20 to get an overview of the semantic regularities of the discourse act of graphic data commentary in the discourse of Economics. The distinctive communicative function of each pattern is also provided.

Table 20
Local grammar patterns of graphic data commentary.

Pattern	Communicative function	Frequency (Normed)
Graphic + Act + Data	Start commenting on research results through general description of the graphic content	406 (413)
Data + Hinge + Graphic	Draw readers' attention to the target graphic and save space for fuller discussion of the graphic data	292 (297)
Graphic + Act + Interpretation	Generalize major findings from a particular graphic	285 (290)
Data + Hinge + Act + Hinge + Graphic	Provide further graphic elaboration for the above-mentioned data	165 (168)
Data + Graphic	Save space for a full discussion of the graphic data	113 (115)
Data + Hinge + Graphic + Hinge + Interpretation	Comment on the value of concrete sets of graphic data	103 (105)
Initiator + Graphic + Data	Start a detailed analysis of the graphic data or present graphic evidence for an argument	95 (97)
Researcher + Act + Data + Hinge + Graphic	Introduce the research process and link the preceding research activities	74 (75)
Data + Hinge + Graphic + Act + Data	Precisely analyze a particular group of data in graphics	65 (66)
Initiator + Graphic + Researcher + Act + Data	Introduce the research process using the graphic data	65 (66)
Data + Hinge + Graphic + Act + Interpretation	Further interpret a particular group of data in graphics	63 (64)
Interpretation + Graphic	Make a full elaboration on the graphic data	48 (49)
Initiator + Graphic + Researcher + Act + Interpretation	Invite readers to interpret the graphic data together	22 (22)
Interpretation + Hinge + Graphic	Identify valuable features of the graphic data through evaluation	22 (22)
Interpretation + Hinge + Act + Hinge + Graphic	Provide an in-depth discussion on the graphic data	16 (16)
Data + Hinge + Graphic + Additional information	Provide reasons to read a particular graphic	11 (11)
Initiator + Graphic + Act + Data	Start a detailed analysis of the graphic data or present graphic evidence for an argument	10 (10)
Total		1,855 (1,886)

As shown in Table 20, in our corpus the discourse act of graphic data commentary manifests 17 high frequency local grammar patterns, corresponding to 1,855 instances or phraseologies (1,886 times per million words). Firstly, the establishment of the local grammar of graphic data commentary provides further evidence that local grammar offers an effective and useful means to analyze language use in ESP/EAP discourse. The characteristic semantic means to conduct a particular discourse act in the discourse of Economics have been systematically represented through local grammar analysis. The 17 local grammar patterns clearly demonstrate how economists communicate meanings to discuss graphic economic data. By performing semantic analysis on the phraseologies of a particular discourse act, local grammar further explores the semantic regularities of the discourse act based on its lexical resources. This fine-grained description of the form-meaning features of discourse acts, we argue, can provide a more useful inventory of language resources for ESP learners. Secondly, the high frequency of graphic data commentary in the corpus suggests that this discourse act is highly frequently performed in research articles of Economics, which supports Swales and Feak's (1994, 1997) view that the commentary of visual data in academic discourse deserves pedagogical attention.

Furthermore, the distinctive discourse features of the 17 local grammar patterns can provide further insights for the investigation of the discourse of Economics. First, the patterns show that giving graphic data commentary in the discourse of Economics involves a limited number of meanings (see Table 2 for all the local grammar terminologies) which can be organized into a limited number of semantic patterns. These local grammar patterns clearly demonstrate what specific meanings can be chosen and how they can be linearly arranged to comment on graphic data. Through the lens of local grammar analysis, economists' preferences of meanings in the phraseologies of graphic data commentary have been precisely described and quantified. This informs that the research of the discourse of Economics can be deepened into the exploration of the semantic regularities of a particular discourse act based on its lexical-grammatical resources. Second, the value of the local grammar patterns also lies in revealing economists' characteristic choices of semantic means to keep coherence and logic of writing and to deal with varying communicative situations. As shown in Table 20, the 17 patterns start with five different meanings, i.e., 'Graphic', 'Data', Interpretation', 'Initiator' and 'Researcher'. The limited five choices allow economists

to flexibly and efficiently choose the most contextually relevant content to discuss, so that they do not break the coherence and logic of writing while commenting on visual data. For example, initiated by 'Researcher', Pattern 17 'Researcher + Act + Data + Hinge + Graphic' enables economists to link the discussion of the graphic data with the unfinished preceding research activity. Local grammar analysis makes it possible to precisely describe what characteristic meanings are available to ensure a coherent and logical commentary of graphic data in the discourse of Economics. On the other hand, as Table 20 shows, each pattern corresponds to a particular communicative function. By matching each pattern with a concrete communicative situation, our local grammar investigation further reveals why a particular semantic choice is made by economists in a specific context. Third, the frequency information of the local grammar patterns shows the typicality of each semantic choice and quantifies economists' preferences in performing graphic data commentary. For example, the top three patterns have a total of 983 occurrences, covering more than 52 % of the instances. Despite the abundance of local grammar patterns, economists tend to rely on an even more limited number of patterns that have been habitually used in their discourse community, which may ensure a quicker recognition and comprehension of the commentaries by the readers. In the meantime, the less frequent patterns (e.g., Pattern 11 in Table 13 which has ten occurrences in the corpus) can provide economists alternative semantic resources to deal with the changing contexts, making their language diversified and flexible. This frequency ranking of the patterns allows a systematic representation of the semantic regularities of a particular discourse act.

7. Pedagogical implications and applications for ESP teaching

The findings of our local grammar investigation can be applicable to ESP/EAP writing pedagogy, especially for developing teaching materials of academic writing in Economics. First, our local grammar analysis informs that teaching academic writing can be concretized into the teaching of discourse acts in writing different parts of academic papers (e.g., claiming importance of research in the introduction section, citing previous research in the literature review section) and the instruction of the semantic regularities of these discourse acts based on local grammar analysis. Second, the key to teaching a discourse act lies in the careful reading of the context to locate an appropriate place for the discourse act and the choice of a most contextually suitable local grammar pattern based on its meaning elements and specific communicative function. This is consistent with our earlier pedagogical practice of applying local grammar in EAP teaching (Zhang and Su, 2021) in which the training of contextual use of local grammar patterns is a key step of the teaching of the discourse act of exemplification. Third, the quantification of local grammar patterns reminds teachers that the teaching of discourse acts needs to be phased (see also Su and Zhang, 2020): the more typical semantic patterns of a discourse act can be prioritized to enable students to cope with general communicative needs; the less typical patterns also deserve teaching as they make students more responsive to constantly changing communicative situations.

To facilitate the application of the above-mentioned pedagogical implications, we propose the following pedagogical steps for teaching discourse acts in academic writing of Economics.

Step 1. Choosing a typical discourse act to teach. Specifically, for the teaching of the writing of a particular section of a research paper in Economics, the teacher works with students to identify a limited number of typical discourse acts in this section by reading authentic research papers. For example, for the writing of Results section, students can be reminded of the typicality and importance of performing the act of graphic data commentary when they see a table or figure is used to display research results.

Step 2. Providing the semantic resources of the target discourse act based on local grammar analysis. In the subsequent step, the local grammar patterns and their corresponding phraseologies of the target discourse act are presented to the students following the frequency order of the patterns. For instance, students are informed that to comment on graphic data, economists most frequently use the phraseologies realizing the semantic pattern of 'Graphic + Act + Data', as in *Table 2 summarizes the target statistics*.

Step 3. Providing lexical resources to facilitate the learning of the patterns. To further assist students' mastery of the patterns, especially specific terminologies in the patterns which have complicated lexical realizations (e.g., the terminology of 'Act' can be realized by dozens of verbs), the teacher provides the typical lexical items realizing such terminologies based on corpus data. For example, for the ten patterns containing 'Act' (e.g., 'Graphic + Act + Data', 'Graphic + Act + Interpretation'), the top ten verbs realizing 'Act' in each pattern are given in Table 21.

Table 21
Top ten verbs realizing 'Act' in each of the ten local grammar patterns.

No.	Pattern	Top ten verbs realizing 'Act'	Example
1	Graphic + Act + Data	show, plot, present, display, illustrate, provide, highlight, graph, reveal, depict	The chart **shows** cumulative net-of-market stock returns for a sample of companies
2	Graphic + Act + Interpretation	show, indicate, suggest, reveal, illustrate, confirm, demonstrate, find, report, imply	Figure 1 **shows** that introductions of intrastate and interstate deregulations are sufficiently independent across states
3	Data + Hinge + Act + Hinge + Graphic		

Table 21 (*continued*)

No.	Pattern	Top ten verbs realizing 'Act'	Example
		report, present, show, define, provide, display, list, summarize, describe, illustrate	Our propensity-score analyses are **reported** in Table 6
4	Researcher + Act + Data + Hinge + Graphic	report, present, re-run, find, re-estimate, plot, provide, estimate, include, replicate	We **report** the CATPCA results in Table 1
5	Data + Hinge + Graphic + Act + Data	show, report, provide, present, indicate, give, summarize, reveal, suggest, reflect	The scores in Table 1 **show** that the mean subjective p(A) is indeed lower than .5 in all five resource
6	Initiator + Graphic + Researcher + Act + Data	provide, present, find, report, estimate, examine, include, measure, test, compare	In Table 1, we **provide** the means, standard deviations, and correlations for the individual difference variables
7	Data + Hinge + Graphic + Act + Interpretation	show, indicate, suggest, demonstrate, reveal, confirm, report, represent, reflect, illustrate	Our correlations in Table 7 **show** that better governed countries have more developed equity markets
8	Initiator + Graphic + Researcher + Act + Interpretation	confirm, see, require, show, note, find[a]	In Panel B of Table 6, we **confirm** the robustness of our finding of a club discount
9	Interpretation + Hinge + Act + Hinge + Graphic	show, describe, report, highlight, depict, suggest	Therefore, nutrition information provision has little influence on consumers' evaluations (as **shown** in Fig. 1 and Fig. 2)
10	Initiator + Graphic + Act + Data	show, indicate, note, reveal	As Fig. 1 **shows**, the declines in stock prices in the US, UK, and Canada were comparable over this period

[a] When 'Act' is realized by less than ten types of verbs in a pattern, all the verb types are provided.

Step 4. Training contextual use of the patterns based on their communicative functions. Since each pattern corresponds to a particular communicative situation, the mastery of it requires the ability to find a suitable pattern for a particular context according to its communicative function. This requires training students' reading of the context to locate a position in which a particular pattern is potentially needed. For instance, if a group of mentioned comparative data needs graphic evidence, students are trained to figure out this need during paper reading and guided to use phraseologies realizing the patterns like 'Graphic + Act + Data' (e.g., *Table 2 shows the comparative data*) or 'Data + Hinge + Act + Hinge + Graphic' (e.g., *The comparative data is provided in Table 2*) to meet this communicative need.

8. Conclusion

This study has provided a local grammar approach for the research of the discourse of Economics and ESP writing pedagogy, focusing on a particular discourse act of graphic data commentary. The fine-grained and context-sensitive local grammar analysis has systematically represented the semantic regularities of graphic data commentary, revealing how a particular discourse act can be realized by a limited number of linear meaning sequences. Specifically, our analysis shows that performing graphic data commentary in the discourse of Economics involves the use of eight meanings (e.g., 'Graphic', 'Data', 'Act'), producing 17 high frequency local grammar patterns. These patterns, highlighting different meanings, suit the discussion of different content in the context and meet varying communicative needs. We might argue that the local grammar approach can provide a more practical means to investigate graphic data commentary and by extension other discourse acts in the discourse of Economics and other discourse types in ESP settings. Pedagogically, we think that such investigation might provide some new insights for teaching academic writing in a particular discipline: starting with discourse acts, providing semantic regularities of the discourse acts based on local grammar analysis, and training contextual use of the semantic regularities. Moreover, we argue that local grammar can be interfaced with corpus-based ESP/EAP genre research to more effectively explore community-specific genre practices. As demonstrated in our study, local grammar enables an in-depth and systematic analysis of the linguistic realization of a particular discourse act from a lexical-semantic perspective. It therefore provides a new possibility to effectively study the linguistic features of the various rhetorical moves in a particular type of ESP/EAP discourse (e.g., the move of presenting research results in the discourse of Economics) by analyzing their characteristic discourse acts or functions (e.g., graphic data commentary).

It has to be acknowledged that this study has limitations. First, as local grammar can only deal with the explicitly expressed phraseologies of graphic data commentary (i.e., phraseologies containing lexical items such as table, figure, etc.), those expressions which are short of explicit lexical items indicating graphic means (e.g., *It also shows opportunities to use somewhat different methods to test those theories*) were not included into our analysis. Second, to what extent local grammar can be applied to describe other discourse acts in the discourse of Economics remains unclear. This is because the instances of some

discourse acts in the discourse of Economics are hard to obtain using corpus method, such as introducing a particular economic phenomenon or testing the robustness of economic research results. Third, although we have discussed the pedagogical insights of local grammar, how can local grammar be integrated into ESP writing courses needs more careful planning since each local grammar is content abundant and the teaching of a local grammar requires extensive classroom time. These developments deserve further analysis in future studies.

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None

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Lei Zhang: Conceptualization, Data curation, Funding acquisition, Methodology, Writing – original draft, Writing – review & editing. **Rui Jiang:** Data curation, Investigation, Writing – review & editing. **Junmei Zhang:** Data curation, Investigation, Writing – review & editing.

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